

petent students of this discipline will have a lot of opportunities in consulting. Similar opportunities will also be created in manufacturing and service sectors," opines Biswajit Mahanty, professor and head, Department of Industrial Engineering and Management, IIT Kharapur.

Like any other industry, in the electronics industry too, there are opportunities in various fields like research and development (R&D), design, engineering, manufacturing, production, procurement and marketing. Each of these fields has its own growth opportunities, more so in design, engineering, production, procurement and marketing.

Engineering management is taught in India under many different titles. One such discipline is industrial engineering and management. Industrial engineering, which deals with optimisation of complex processes or systems, is also known as operations management, management science, operations research, systems engineering or manufacturing engineering—usually, depending on the sub-specialties involved.

"American Institute of Industrial Engineers (AIIE) and all other professional bodies across the world have defined industrial engineering as concerned with the design, improvement and installation of integrated systems of men, materials and equipment. It draws upon specialised knowledge and skill in the mathematical, physical and social sciences together with the principles and methods of engineering analysis and design, to specify, predict and evaluate the results to be obtained from such systems," says Dr R.P. Mohanty, former chairman, Indian Institution of Industrial Engineering (IIIE).

Opportunities in the electronics manufacturing and PCB assembly industries are very good—

Suresh Nair, director, Leaptech Corporation

“There is a need for technical people who are also futuristic and have leadership skills in the industry— Prof. M.H. Bala Subrahmanya, chairman, Department of Management Studies, IISc



Who should opt for it?

Before you embark on this journey that brings together the problem-solving skills of engineering and the organisational abilities of management to oversee complex enterprises from conception to completion, it is important to evaluate whether this career is a good fit for you.

Candidates who wish to pursue engineering management as profession should be able to solve challenging problems, visualise solutions using computer modeling and design tools, and apply abstract thought to problem solving. These skills become important as engineering managers are required to oversee the design of machinery, equipment, products or systems; design and assess the feasibility of new products or processes; oversee direct production, quality assurance or maintenance; and coordinate with other units such as management, financial and marketing.

What role to choose?

A degree in engineering management qualifies you for a diverse array of jobs—one of the most attractive facets of this field. These positions range from the academic to the technical. As an engineering manager, you will have to focus on product development, materials management, production processes and workforce reliability.

The career paths for a professional in this field may include manufacturing, financial services, logistics,

engineering, technical sales, project planning/management and quality management/control. Following are a few job titles for engineering management professionals:

1. Project manager
2. Manufacturing engineer
3. Consultant
4. Quality assurance engineer
5. Financial analyst
6. Production supervisor
7. Systems engineer and analyst
8. Technical sales representative

As an engineering manager, your responsibilities will largely be in staff position rather than in line position. "This means the engineering management graduates will more often than not support the top management rather than become part of the top management themselves," adds Prof. Mahanty.

How to get started?

Manipal Institute of Technology (MIT) is one of the few institutes in India that offers M.Tech in engineering management. It is the first PG (two-year) course that MIT started offering in 1988.

"Most of the students who approach us for this course apply after a few years of experience in the technical field. They are taught subjects like resource optimisation techniques in engineering concerns and production and operations management including modeling and simulation techniques using system dynamics in supply chain management both in production and service sectors. Students take up industry-related projects in their second year. They select a problem in an industry of their choice and present a dissertation on the problem, analysis and solution. This involves academic

